

Generation Interconnection System Impact Study Report

For Queue Project Q13-0249

Lakewood 138 kV

May 19, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0249. The Interconnection Customer proposes a 550.3 MW natural gas combined cycle facility interconnecting at the Lakewood 138 kV substation in the New Jersey area. The request was received on June 29, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 147 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,362,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WEBSTER RD to SOUTHEAST ATE 138.0 kV
Contingency: LEISURE VILLAGE to CAMPO FRESCO 2 138.0 kV
Loading: 230.8% (Rating: 160 MVA)
DFAX: 58.9%
Pre-project loading: 28.1%
2. Facility: LAKEWOOD to SOUTHEAST GATE 138.0 kV line
Contingency: LEISURE VILLAGE to CAMPO FRESCO 2 138.0 kV line
Loading: 183.7% (Rating: 211 MVA)
DFAX: 58.9%
Pre-project loading: 30.2%
3. Facility: LAKEWOOD to CAMPO FRESCO 2 138.0 kV
Contingency: LAKEWOOD to SOUTHEAST GATE 138.0 kV L
Loading: 144.3% (Rating: 232 MVA)
DFAX: 65.6%
Pre-project loading: 11.3%
4. Facility: LEISURE VILLAGE to CAMPO FRESCO 2 138.0 kV L
Contingency: LAKEWOOD to SOUTHEAST GATE 138.0 kV
Loading: 131.3% (Rating: 270 MVA)
DFAX: 50.1%
Pre-project loading: 29.3%
5. Facility: OXY CENTURY to LAKEWOOD 138.0 kV
Contingency: CAMPO FRESCO 2 to LAKEWOOD 138.0 kV
Loading: 120.6% (Rating: 189 MVA)
DFAX: 49.5%
Pre-project loading: 23.4%
6. Facility: LEISURE VILLAGE to PECAN GROVE 138.0 kV
Contingency: LAKEWOOD to SOUTHEAST GATE 138.0 kV
Loading: 111.1% (Rating: 178 MVA)

DFAX: 36.0%
Pre-project loading: 0.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WEBSTER RD to SOUTHEAST ATE 138.0 kV	Rebuild line with higher-capacity conductor	\$2,056,000	100% (\$2,056,000)
LAKEWOOD to SOUTHEAST GATE 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,296,000	100% (\$1,296,000)
LAKEWOOD to CAMPO FRESCO 2 138.0 kV	Rebuild line with higher-capacity conductor	\$1,563,000	100% (\$1,563,000)
CAMPO FRESCO 2 to LEISURE VILLAGE 138.0 kV line	Rebuild line with higher-capacity conductor	\$927,000	100% (\$927,000)
LAKEWOOD to OXY CETURY 138.0 kV	Rebuild line with higher-capacity conductor	\$827,000	100% (\$827,000)
LEISURE VILLAGE to PECAN GROVE 138.0 kV	Rebuild line with higher-capacity conductor	\$1,693,000	100% (\$1,693,000)
	Total allocated to project		\$8,362,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.